U.S. DEPARTMENT OF ENERGY'S
CYBERFORCE® PROGRAM

CyberForce® 101

Anomalies

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CyberForce Competition® Overview

The CyberForce Competition® is an in-person competition that provides collegiate students with a cyber-physical hands-on, realistic energy-based scenario to secure and defend that incorporates life-like constraints and tasks, actual end users, and continual emerging threats. Students are challenged by the opportunity to apply classroom knowledge in a real-life practical environment.

The goal is to build the pipeline of qualified cybersecurity candidates for crucial positions such as those in our energy sector, national laboratories, and industry. The CyberForce Competition provides an inclusive space for all participants, including collegiate students, mentors, industry, and academia.

Anomalies Overview

Anomaly scoring is one of the five scored areas within the CyberForce Competition®. Anomalies are designed to simulate the real-world, potentially time-sensitive tasks faced by working professionals. These competition challenges test both technical and non-technical knowledge, critical thinking skills, and time management of the cyber professionals.

What are Anomalies

Anomalies are crafted challenges that are mapped to the NIST NICE Framework and include tasks that are meant to pull attention in various directions and require teams to strategize, coordinate, and prioritize their team's direction. Anomalies are created and deployed to achieve: 1) exposing students to real world challenge that may occur within the field of information

and operational technology; 2) support the Red team with added exploits as needed; 3) measure students' ability to solve a variety of technical and non-technical problems; 4) simulate a real-world network where administrators have to balance several tasks throughout their day; and 5) provide coordination roles and prioritization opportunities for teams to identify their strengths amongst their team.

Scoring and Syntax

Each anomaly has a specific number of attempts allowed, which varies based on difficulty. Trivia/easy-based anomalies allow for only one attempt and as the difficulty increases, the number of attempts does as well. The difficulty of the anomaly also depends on the number of points awarded per correct answer. Most anomalies are not case-sensitive but are syntax-sensitive, meaning proper spelling, grammar, and the use of special characters are the key. If an anomaly is case-sensitive, it will be explicitly mentioned in the question. Pay close attention to syntax hints if provided in the question area.

Examples

Below are some examples of previous anomalies. As shown, some are technical in nature, and some require you to get up and look around the room for a clue or flag.

Snort Anomaly	Analyze a packet capture file to identify the first instance of malware based on specific traffic patterns.
Wireshark Anomaly	Examine a packet capture to determine the image opened from a potentially malicious website.
Find Breaker	BREAKER BREAKER, Come in BREAKER! Oh no! We lost Breaker!!! He's gone back in TIME and needs to be found! Once you find him, enter the flag found within the brackets.

Preparation Tips

Individuals can become familiar with tools and software such as Wireshark, NMAP, Steghide, Python, etc. Additionally, practicing in other open capture the flags (CTFs) may be beneficial.

- Conquer the Hill Series
  - https://cyberforce.energy.gov/conquer-the-hill/
- picoCTF
  - https://picoctf.org/
- Hack the Box
  - https://www.hackthebox.com/hacker/ctf